

■ 3.4.5 Advanced Topic: Existence of Solutions

Using **Reduce**, you can find out under exactly what conditions a particular set of equations has solutions. **Solve** tells you whether any generic solutions exist.

There is no value of x which solves these simultaneous equations. **Reduce** thus simplifies the logical statement $x==1 \ \&\& \ x==2$ to the explicit value **False**.

```
In[1]:= Reduce[ x == 1 && x == 2 , x ]
Out[1]= False
```

There is a solution to these equations, but only when a has the special value 1.

```
In[2]:= Reduce[ x == 1 && x == a , x ]
Out[2]= a == 1 && x == 1
```

The solution is not generic, and is rejected by **Solve**.

```
In[3]:= Solve[ x == 1 && x == a , x ]
Out[3]= {}
```

This equation is true for any value of x .

```
In[4]:= Reduce[ x == x , x ]
Out[4]= True
```

This is the kind of result **Solve** returns when you give an equation that is always true.

```
In[5]:= Solve[ x == x , x ]
Out[5]= {{}}
```

When you work with systems of linear equations, you can use **Solve** to get generic solutions, and **Reduce** to find out for what values of parameters solutions exist.

Here is a matrix whose i, j^{th} element is $i + j$.

```
In[6]:= m = Table[i + j, {i, 3}, {j, 3}]
Out[6]= {{2, 3, 4}, {3, 4, 5}, {4, 5, 6}}
```

The matrix has determinant zero.

```
In[7]:= Det[ m ]
Out[7]= 0
```

This makes a set of three simultaneous equations.

```
In[8]:= eqn = m . {x, y, z} == {a, b, c}
Out[8]= {2 x + 3 y + 4 z, 3 x + 4 y + 5 z,
         4 x + 5 y + 6 z} == {a, b, c}
```

Solve reports that there are no generic solutions.

```
In[9]:= Solve[eqn, {x, y, z}]
Out[9]= {}
```

Reduce, however, shows that there *would* be a solution if the parameters satisfied the special condition $c == -a + 2b$.

```
In[10]:= Reduce[eqn, {x, y, z}]
Out[10]= c == -a + 2 b && x == -4 a + 3 b + z &&
         y == 3 a - 2 b - 2 z
```

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

© 1988 Wolfram Research, Inc. Permission is hereby granted for web users to make one paper copy of this page for their personal use. Further reproduction, or any copying of machine-readable files (including this one) to any server computer, is strictly prohibited.

For nonlinear equations, the conditions for the existence of solutions may be very complicated.

Here is a very simple pair of nonlinear equations.

```
In[11]:= eqn = {x y == a, x^2 y^2 == b}
```

```
Out[11]= {x y == a, x^2 y^2 == b}
```

Solve shows that the equations have no generic solutions.

```
In[12]:= Solve[eqn, {x, y}]
```

```
Out[12]= {}
```

Reduce gives the complete conditions for a solution to exist.

```
In[13]:= Reduce[eqn, {x, y}]
```

```
Out[13]= y != 0 && b == a^2 && x == a/y ||
a == 0 && b == 0 && y == 0
```